

Abhishek Chaudhary

Summary

AI Engineer with hands-on experience building **LLM-powered, multimodal, and GenAI systems**. Strong background in transformer fine-tuning, RAG pipelines, document intelligence, and production-grade deployment across healthcare and travel domains using PyTorch, Hugging Face, and cloud-based inference.

Technical Skills

Languages: Python, Java, C/C++, JavaScript

LLMs & GenAI: OpenAI, Gemini, Llama 3.x, Qwen 2.5, Prompt Engineering, Instruction Tuning, RAG

ML / NLP: Transformers, Text Summarization, NER, Tokenization, Model Evaluation (ROUGE, Accuracy)

Computer Vision: YOLO, LayoutLMv3, PaddleOCR, Multimodal Document AI

Frameworks & MLOps: PyTorch, Hugging Face, LangChain, FastAPI, Docker, AWS

Databases & Tools: MongoDB, MySQL, Git, GitHub, Postman, Jupyter

Experience

AI Engineer

Dec 2024 – Oct 2025

Digiquanta / Mondee, Hyderabad

- Designed and deployed **LLM-driven autonomous travel agents** for flight/hotel search, itinerary generation, and personalized recommendations.
- Built **multimodal medical document intelligence pipelines** using Qwen 2.5-VL and Llama 3.1, automating **95%** of manual data extraction.
- Fine-tuned **LayoutLMv3** for document classification and field extraction, achieving **98% accuracy** and reducing QA effort by **90%**.
- Trained and optimized **YOLO-based vision models** for handwritten signature detection with **95% precision** on scanned medical forms.
- Created a patient risk scoring and rehospitalization prediction models, reaching **75% predictive accuracy**.
- Implemented **RAG pipelines** with semantic chunking and embedding-based retrieval to improve factual accuracy of LLM responses.
- Implemented a **multilingual AI medical assistant** (English, Hindi, Telugu) with sub-**10s** inference latency using OpenAI and Gemini APIs.
- Improved flight and hotel price prediction accuracy by **30%** through feature engineering and ensemble learning.

Backend Developer Intern

Jan 2024 – Jun 2024

Enthuons Technologies Pvt. Ltd., Noida

- Developed and deployed **12+ RESTful APIs** using Node.js and Express.js, supporting **1,000+ requests/min**.
- Optimized database queries, middleware, and error handling, reducing backend failures by **20%**.
- Collaborated in Agile sprints, contributing to code reviews, debugging, and version control using Git and GitHub.

Projects

AI Comedy Video Generator

Aug 2024

- Built an end-to-end GenAI pipeline for script generation, avatar-based video synthesis, and TTS using Gemini and D-ID.
- Developed a React-based frontend with Node.js backend to deliver automated, multilingual AI-generated video content.

Transformer-based Text Summarization System

Jul 2024

- Fine-tuned transformer models on **15K+ dialogue samples** using PyTorch and Hugging Face.
- Improved summary coherence and relevance by **20% ROUGE score** through preprocessing and hyperparameter tuning.

Education

B.Tech, Mechanical Engineering

2020 – 2024

Delhi Technological University (DTU), Delhi — CGPA: 7.6 / 10

Achievements

- Finalist, **Code Gladiators 2024** – Rank 191 among 35,000+ participants
- Solved **700+** DSA problems (CodeChef 1619, Codeforces 1286)